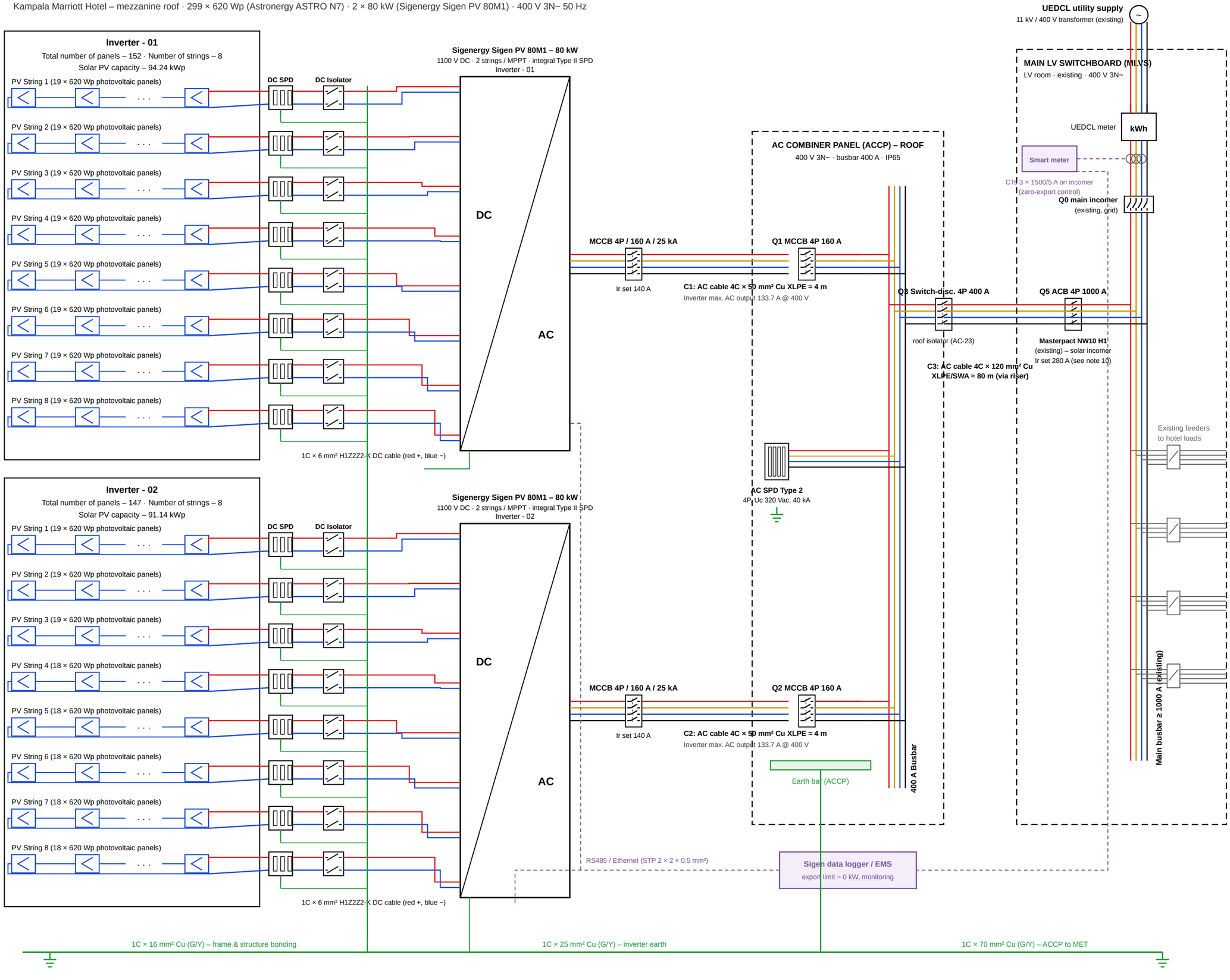
SINGLE LINE DIAGRAM – 185.4 kWp GRID-TIED ROOFTOP SOLAR PV SYSTEM (ZERO EXPORT)

Kampala Marriott Hotel – mezzanine roof · 299 × 620 Wp (Astronergy ASTRO N7) · 2 × 80 kW (Sigenergy Sigen PV 80M1) · 400 V 3N~ 50 Hz



General Notes

This diagram shows the electrical schematic of the 185.4 kWp grid-tied solar PV system (zero export, no battery).

1. **PV modules**
   Astronergy ASTRO N7 CHSM66RN(DG)/F-BH, 620 Wp bifacial; Voc 49.04 V, Isc 16.11 A, Vmp 41.56 V, Imp 14.92 A
2. **Strings**
   INV-01: 8 × 19 = 152 (94.24 kWp); INV-02: 3 × 19 + 5 × 18 = 147 (91.14 kWp). Max string Voc ≈ 955 V (15 °C) < 1100 V
3. **Inverters**
   2 × Sigenergy Sigen PV 80M1, 80 kW, 400 V 3N~; max AC current 133.7 A; 2 strings per MPPT (max 40 A/MPPT)
4. **DC cables**
   1C × 6 mm² Cu H1Z2Z2-K, 1.5 kV DC, IEC 62930 / EN 50618, UV resistant
5. **AC cables**
   C1, C2: 4C × 50 mm² Cu XLPE; C3: 4C × 120 mm² Cu XLPE/SWA, 0.6/1 kV, IEC 60502-1; VD < 2 %
6. **Earthing cables**
   16 / 25 / 70 mm² Cu G/Y, 450/750 V, IEC 60364-5-54
7. **DC SPD (per string)**
   Type 2, Ucpv 1100 V DC, In 20 kA, Imax 40 kA, Up ≤ 4.0 kV, IEC 61643-31
8. **DC isolator (per string)**
   2-pole, 1200 V DC, 32 A, DC-PV2, IEC 60947-3, IP66 enclosure
9. **AC SPD (ACCP)**
   Type 2, 4P (3P+N), Uc 320 Vac, In 20 kA, Imax 40 kA, Up ≤ 1.5 kV, IEC 61643-11
10. **AC protection**
    Inverter MCCB 4P 160 A (Ir 140 A); ACCP Q1/Q2 MCCB 4P 160 A (Ir 140 A); Q3 switch-disconnector 4P 400 A. Q5 = existing Masterpact NW10 H1 ACB used as solar incomer: fit 400 A sensor/rating plug and set Ir 280 A to protect C3 (4C × 120 mm²). If the 1000 A sensor is kept (min Ir 400 A), upsize C3 to 4C × 185 mm².
11. **Zero export**
    Smart meter + 3 CTs on MLVS incomer → Sigen EMS limits inverter output to keep export = 0
12. **Solar system**
    PV capacity 185.38 kWp DC; inverter capacity 2 × 80 kW = 160 kW AC; DC/AC 1.16
13.
    Cable lengths indicative; confirm on site. Verify MLVS spare way, fault level and busbar rating before installation.

Legend

- DC + / L1
- DC − / L3
- L2
- N
- Earth (PE)
- Comms / CT signal

Firm name & address	**NileSun Power Ltd, Kampala, Uganda**
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